Takara Sakai

Specially Appointed Assistant Professor · School of Environment and Society · Institute of Science Tokyo (Takayama Lab)

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RESEARCH INTERESTS

- Transportation and Traffic Theory
- Regional Science and Urban Economics
- Optimization, Control & Algorithms
- Game Theory & Mechanism Design

ACADEMIC APPOINTMENTS

Specially Appointed Assistant Professor, School of Environment and Society, Institute of Science Tokyo	**Oct. 2024 – present**
Specially Appointed Assistant Professor, School of Environment and Society, Tokyo Institute of Technology	**Oct. 2023 – Sep. 2024**
Research Fellow (DC1), Japan Society for the Promotion of Science (JSPS)	**Apr. 2020 – Sep. 2023**

EDUCATION

Ph.D. in Information Science, Tohoku University Advisor: Prof. Takashi Akamatsu	**2023**
M.S. in Information Science, Tohoku University	**2020**
B.S. in Civil Engineering, Tohoku University	**2018**

REFEREED JOURNAL ARTICLES

1. **Takara Sakai**, Takashi Akamatsu, Koki Satsukawa (2026). Queue Replacement Approach to Dynamic User Equilibrium Assignment with Route and Departure Time Choice. *Transportation Research Part B: Methodological*, 211, 103523. doi:10.1016/j.trb.2026.103523

2. **Takara Sakai**, Yuki Takayama (2025). Demand-driven transport network formation: Emergence of a hub-and-spoke structure (in Japanese). *Journal of JSCE*, 81(6), ID:24-00167. doi:10.2208/jscejj.24-00167

3. **Takara Sakai**, Takashi Akamatsu, Koki Satsukawa (2024). Queue replacement principle for corridor problems with heterogeneous commuters. *Transportation Research Part B: Methodological*, 187, 103024. doi:10.1016/j.trb.2024.103024

4. **Takara Sakai**, Takashi Akamatsu, Koki Satsukawa (2024). A Paradox of Telecommuting and Staggered Work Hours in the Bottleneck Model. *Transportation Science*, 58(6), 1335–1351. doi:10.1287/trsc.2024.0520

5. **Takara Sakai**, Masanao Wakui, Takashi Akamatsu (2023). An efficient solution method for the Fujita–Ogawa model in ultra-large-scale discrete space (in Japanese). *Journal of JSCE*, 79(4).

6. Masanao Wakui, **Takara Sakai**, Takashi Akamatsu (2023). An efficient solution method for dynamic user equilibrium assignment in large-scale networks (in Japanese). *Journal of JSCE*, 79(4).

7. **Takara Sakai**, Takashi Akamatsu, Koki Satsukawa (2021). Departure time choice problem in tandem bottlenecks with heterogeneous schedule cost (in Japanese). *Journal of JSCE, Ser. D3 (Infrastructure Planning and Management)*, 77(4), 330–345. doi:10.2208/jscejipm.77.4_330

8. Takashi Akamatsu, Takeshi Nagae, Minoru Osawa, Koki Satsukawa, **Takara Sakai**, Daijiro Mizutani (2021). Model-based analysis on social acceptability and feasibility of a focused protection strategy against the COVID-19 pandemic. *Scientific Reports*, 11, 2003. doi:10.1038/s41598-021-81630-9

9. **Takara Sakai**, Takashi Akamatsu (2019). Congestion patterns and stability of the Macroscopic Fundamental Diagram in the Tokyo metropolitan expressway network (in Japanese). *Journal of JSCE, Ser. D3 (Infrastructure Planning and Management)*, 75(2), 97–108. doi:10.2208/jscejipm.75.97

WORKING PAPERS & PREPRINTS

1. **Takara Sakai**, Riki Kawase (2026). Coarse Preference Reporting in the Bottleneck Model: Approximate Strategyproofness and Efficiency. arXiv:2606.17400
2. Mingzhi Xiao, **Takara Sakai**, Daisuke Murakami, Yuki Takayama (2025). When Does Tourism Raise Land Prices? Threshold Effects, Superstar Cities, and Policy Lessons from Japan. arXiv:2509.04307
3. Shota Fujishima, **Takara Sakai**, Yuki Takayama (2025). Quantifying Congestion Externalities in Road Networks: A structural estimation approach using a stochastic evolutionary model. RIETI Discussion Paper 25-E-055
4. **Takara Sakai**, Koki Satsukawa, Takashi Akamatsu (2022). Non-existence of queues for system optimal departure patterns in tree networks. arXiv:2205.06015

SELECTED INTERNATIONAL CONFERENCE PRESENTATIONS

1. **Takara Sakai**, Takashi Akamatsu, Koki Satsukawa. Queue Replacement Approach to Dynamic User Equilibrium Assignment with Route and Departure Time Choice. *26th International Symposium on Transportation and Traffic Theory (ISTTT)*, Munich, Germany, Jul. 2026.

2. **Takara Sakai**, Yasunari Hikima. Inverse Optimal Transport for Bottleneck Models: Estimating Temporal Travel Demand Distributions. *INFORMS Transportation and Logistics Society Third Triennial Conference (TSL)*, MIT, MA, USA, Jul. 2026.

3. Mingzhi Xiao, **Takara Sakai**, Yuki Takayama. High-Speed Rail and Population Agglomeration in China: Implications for Urban Planning and Regional Development. *World Conference on Transport Research (WCTR)*, Toulouse, France, Jul. 2026.

4. Mingzhi Xiao, Yuki Takayama, **Takara Sakai**. The Role of High-Speed Rail in Shaping Chinese County-Level Economic Structures. *HKSTS International Conference*, Hong Kong, Dec. 2025.

5. **Takara Sakai**, Takashi Akamatsu, Koki Satsukawa. A Paradox of Telecommuting and Staggered Work Hours in the Bottleneck Model. *25th International Symposium on Transportation and Traffic Theory (ISTTT)*, Ann Arbor, MI, USA, Jul. 2024.

6. **Takara Sakai**. A Paradox of Telecommuting and Staggered Work Hours in the Bottleneck Model. *International Seminar, Huazhong University of Science and Technology*, Wuhan, China, Mar. 2024.

7. **Takara Sakai**, Takashi Akamatsu, Koki Satsukawa. Welfare impacts of remote and flexible working policies in the bottleneck model. *11th Symposium of the European Association for Research in Transportation (hEART)*, ETH Zurich, Sep. 2023.

8. **Takara Sakai**, Takashi Akamatsu, Koki Satsukawa. Queue replacement principle for corridor problems with heterogeneous commuters. *9th International Symposium on Dynamic Traffic Assignment (DTA)*, Northwestern University, USA, Jul. 2023.

9. **Takara Sakai**, Takashi Akamatsu, Koki Satsukawa. Departure time choice problems in a corridor network with heterogeneous value of schedule delay. *25th HKSTS International Conference*, online, Dec. 2021.

10. **Takara Sakai**, Takashi Akamatsu. Spatio-Temporal Regularities of Traffic Congestion Patterns in the Metropolitan Expressway Network. *5th CWRU–Tohoku Joint Workshop*, Tohoku University, Japan, Aug. 2018.

GRANTS & FELLOWSHIPS

1. Reconstruction of dynamic user equilibrium assignment theory based on the queue replacement principle.
   JSPS KAKENHI, Grant-in-Aid for Early-Career Scientists (No. 26K17486) · Apr. 2026 – Mar. 2029 · Principal Investigator

2. Mathematical analysis of road traffic network systems with mixed cooperative and non-cooperative vehicles.
   Obayashi Foundation, Research Grant · Apr. 2026 – Mar. 2027 · Principal Investigator

3. Assistant Placement Program.
   Institute of Science Tokyo, DE&I Division · Apr. 2025 – Sep. 2026 · Principal Investigator

4. Stability analysis of transport systems using Max-Plus algebra.
   Institute of Science Tokyo, Academy for Broad-based Basic Research, New Research Development Incentive Grant · Sep. 2024 – Mar. 2026 · Co-Investigator (PI: Riki Kawase)

5. Development of a traffic network equilibrium model capable of representing and manipulating information- and strategy-sharing among vehicles.
   JSPS KAKENHI, Grant-in-Aid for Research Activity Start-up (No. 24K22973) · Aug. 2024 – Mar. 2027 · Principal Investigator

6. Complementary and conflicting effects of commuting congestion mitigation measures at different spatio-temporal scales.
   Obayashi Foundation, Research Grant · Apr. 2024 – Mar. 2025 · Principal Investigator

7. Traffic control utilizing congestion pattern information in urban expressway networks.
   JSPS KAKENHI, Grant-in-Aid for JSPS Fellows (No. 20J21744) · Apr. 2020 – Mar. 2023 · Principal Investigator

Last updated: July 2026